

Assessment of migrating endocrine-disrupting chemicals in bottled acidic juice using type UVM-7 mesoporous silica modified with cyclodextrin

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ABSTRACT

An innovative material based in type UVM-7 mesoporous silica containing analyte-accessible cyclodextrin units is described and assessed as a selectivity-enhancing sorbent for extracting endocrine-disrupting chemicals from bottled apple juice to subsequently quantify them. The synthesis procedure has been carried out using both β - and γ -cyclodextrin for their later comparison. Then, a complete analytical method for the isolation and determination of the above analytes has been validated. Following the optimized procedure, recoveries between 94% and 100% have been achieved and good repeatability is obtained with deviations under 6.8% for intra-day and inter-day experiments. The detection limits of the method have been established in the ng L⁻¹ level, which demonstrates the ability to quantify the trace concentrations established by sanitary restrictions. A low matrix effect is found when working with real samples. To end, a comparison with an alternative extraction method using C₁₈ extraction cartridges has been carried out and the results obtained with both procedures are comparable.

KEYWORDS

Mesoporous silica materials, UVM-7, Cyclodextrin, Endocrine-disrupting chemicals, Apple juice

1	ABBREVIATIONS
2	ACN, acetonitrile
3	BPA, bisphenol A
4	BPAP, bisphenol AP
5	BPC, bisphenol C
6	CD, cyclodextrin
7	CTAB, cetyl trimethylammonium
8	EDC, endocrine-disrupting chemical
9	EP, ethylparaben
10	EPA, <i>Environmental Protection Agency</i>
11	HPLC-FD, high-performance liquid chromatography coupled to fluorescence detection
12	LOD, limit of detection
13	LOQ, limit of quantification
14	MeOH, methanol
15	MP, methylparaben
16	NMR, nuclear magnetic resonance
17	PC, polycarbonate
18	PE, polyethylene
19	PET, polyethylene terephthalate
20	PP, propylparaben
21	PVC, polyvinyl chloride
22	REACH, EU regulation from <i>Registration, Evaluation, Authorization, and restriction of Chemicals</i>
23	RSD, relative standard deviation
24	SPE, solid-phase extraction
25	TEA, triethanolamine
26	TEM, transmission electron microscopy
27	TEOS, tetraethyl orthosilicate
28	TGA, thermogravimetric analysis
29	XRD, X-ray diffraction

1. INTRODUCTION

Synthetic resins are widely used to bestow some plastic properties (Bang et al., 2012) such as flame resistance, color, and softness. Polyethylene (PE), polyvinyl chloride (PVC), polyethylene terephthalate (PET), or polycarbonate (PC) can be named among the major ingredients for the manufacture of an extensive variety of goods (Bang et al., 2012; Khan, Alammari, Aqel, & Azam, 2020). However, they are a source of concern due to the presence of endocrine-disrupting substances in their structure, which may pose some risk to the environment and human health (Bang et al., 2012). Since they are not chemically bonded therein, unintentional contamination of endocrine-disrupting chemicals in food coming from plastic containers may happen easily, especially under specific conditions such as high temperature, acidic pH, and sunlight (Arfaeinia et al., 2020; Muncke, 2009).

Endocrine disruption is known as the disruption of the hormonal and physiological balance of the body as a result of natural or synthetic chemicals interfering with endocrine pathways (Kaya, Cetinkaya, Bakirhan, & Ozkan, 2020). Potential health concerns include diabetes, thyroid dysfunction, reproductive disorders, developmental defects, and cancer. Thus, several chemical substances have been identified as endocrine-disrupting chemicals (EDCs). Some common examples are alkylphenols and bisphenols, phthalates, parabens, pesticides, hormones, polychlorinated biphenyls, or heavy metals (Chormey, Zaman, Kasa, & Bakirdere, 2020), among others. As a matter of fact, most of them are included in the 2021 European Union's REACH (1907/2006/EC, 2006) SVHC (Substances of Very High Concern) list and legislative regulations about their maximum concentrations allowed by the European Commission (2018/213/EU, 2018). The EPA has also established reference doses for them. Among them, bisphenols attract special attention as ubiquitous substances used to make food and beverage containers and as additives (Khan et al., 2020). Concretely, bisphenol A (BPA) triggers high concern and has been determined in beverages coming from cans and plastic bottles, for example (Bang et al., 2012; Khan et al., 2020; Wagner & Oehlmann, 2009). Besides, parabens are popular synthetic preservatives used not only in food products but also in pharmaceutical and cosmetic preparations. Our daily exposure to them has been estimated as significant, with 25 mg coming

1 from pharmaceuticals, 50 mg from cosmetics, and 2.5 mg from food products (Matwiejczuk,
2 Galicka, & Brzóška, 2020).

3 Due to the existing variety of EDCs, our lifestyle means that we are in constant contact with
4 them. The strict regulations that require trace detection due to the lack of certain threshold values
5 for adverse effects and their low concentrations in products make it necessary to develop
6 sensitive monitoring methods (Kaya et al., 2020). A variety of procedures has been reported.
7 They consist mostly of extraction methodologies (Azzouz & Ballesteros, 2016; Chormey et al.,
8 2020) including solid-phase extraction (SPE) (Azzouz, Rascón, & Ballesteros, 2016a; Brede,
9 Fjeldal, Skjevrak, & Herikstad, 2003), liquid-liquid extraction (Asimakopoulos, Wang, Thomaidis,
10 & Kannan, 2014), pressurized liquid extraction (Ferrer et al., 2011), solid-phase microextraction
11 (Mousa, Basheer, & Rahman Al-Arfaj, 2013), stir-bar sorptive extraction (Chang, Shen, Yang, &
12 Wu, 2012), and liquid-liquid microextraction (Vela-Soria et al., 2013) mostly coupled to
13 chromatographic detection (Kaya et al., 2020). Among them, SPE is the preferred choice for
14 cleaning and preconcentrating these substances due to its suitability to the analysis of compounds
15 of different polarity and properties. For this reason, some authors have used it to preconcentrate
16 not only EDCs but also other substances in complex matrices (Azzouz, Rascón, & Ballesteros,
17 2016b). However, using commercial solid phases presents some limitations (Belenguer-Sapiña,
18 Pellicer-Castell, Mauri-Aucejo, Simó-Alfonso, & Amorós, 2021) such as the lack of specificity.
19 Besides, the simultaneous presence of polar groups may complicate the quantitation step of
20 EDCs in certain determination routes (Azzouz et al., 2018) to require additional derivatization.
21 Due to the need to use these extraction strategies, there is a special interest in developing new
22 sorbent materials for the selective retention of analytes (Pellicer-Castell, Belenguer-Sapiña,
23 Amorós, El Haskouri, et al., 2020).

24 The use of self-designed nanostructured materials such as metal-organic frameworks or
25 molecularly imprinted polymers in sorptive techniques has thus been recognized as a potential
26 sample pretreatment step to ease the isolation of the analytes of interest (Azzouz et al., 2016b).
27 Particularly interesting are host-guest complexing paths, which are already being exploited for the
28 reversible adsorption of substances and can provide an improvement in some features such as

selectivity (Belenguer-Sapiña et al., 2021). Cyclodextrins (CDs) are in this sense a promising tool due to their capability of encapsulating suitable analytes based on their polarity and physical properties (Belenguer-Sapiña, Pellicer-Castell, Amorós, Simó-Alfonso, & Mauri-Aucejo, 2020).

As structuring material, mesoporous silica offers remarkable features as sorbent such as high stability, large surface areas, and the possibility of adjusting the pore size (El Haskouri et al., 2008; Pellicer-Castell, Belenguer-Sapiña, Amorós, El Haskouri, et al., 2020). Thus, UVM-7 silica can be considered as a nanometric version of the MCM-41 materials (Pellicer-Castell et al., 2019), being its main feature the particular architecture shown. It consists of a continuous network constructed from aggregated small mesoporous nanoparticles generating a non-ordered accessible pore system with a high surface area (Pellicer-Castell, Belenguer-Sapiña, Amorós, Herrero-Martínez, & Mauri-Aucejo, 2020). Some applications of UVM-7 materials have been reported (Martínez Pérez-Cejuela et al., 2018; Pellicer-Castell et al., 2018, 2019). However, their modification with cyclodextrins as selectivity-enhancing units and their subsequent application to the analysis of high-concern substances have been barely studied.

This work aimed to assess the utility of innovative type UVM-7 mesoporous silica modified with cyclodextrin (*Figure S1*) as SPE sorbent for the extraction of EDCs in widely consumed fruit juices. Since apple juices are characterized for showing a high acidic pH and are usually packaged in PET plastic containers while stored for a long time, the presence of EDCs in these liquids may be expected (Arfaeina et al., 2020). To this, the relevant analytical features of the method were established. The results were compared with those obtained using a reference extraction method reported in the literature (Khan et al., 2020).

2. MATERIAL AND METHODS

2.1. Reagents

On the one hand, the individual bisphenol A (BPA), bisphenol AP (BPAP), and bisphenol C (BPC) analytical standards were purchased from Sigma-Aldrich (www.sigmaaldrich.com, St. Louis, USA). On the other hand, methylparaben (MP), ethylparaben (EP), and propylparaben (PP) $\geq 99\%$ were acquired from Fluka (www.lab-honeywell.com, Bucharest, Romania). The

1 concentrated individual standard solutions were prepared in methanol and the working standard
2 multicomponent solutions were obtained by their dilution and stored at -18 °C. HPLC grade
3 methanol (MeOH) and acetonitrile (ACN) were acquired in VWR ProLabo Chemicals
4 (www.vwr.com, Radnor, USA). In addition, ultrapure water from an Adrona (www.adrona.lv, Riga,
5 Latvia) purification system was employed during the experimental procedure. Other reagents
6 such as NaCl (s) $\geq 99.5\%$, NaOH (s) $\geq 98.5\%$, and H₂SO₄ (aq) 98% were bought from Panreac
7 AppliChem (www.itwreagents.com, Barcelona, Spain).

8 For the synthesis of the solid phase, a first functionalization step for the cyclodextrins used
9 was required. To it, β - and γ -cyclodextrin were purchased from CycloLab (www.cyclolab.hu,
10 Budapest, Hungary), and 3-(triethoxysilyl)propyl isocyanate and dry pyridine $\geq 99.5\%$ were
11 acquired from Sigma-Aldrich (www.sigmaaldrich.com, St. Louis, USA). The obtention of
12 mesoporous silica materials was carried out using cetyl trimethylammonium (CTAB) reagent
13 grade from Sigma-Aldrich (web, St. Louis, USA), ethanol from VWR ProLabo Chemicals
14 (www.vwr.com, Radnor, USA), as well as tetraethyl orthosilicate $\geq 98\%$ (TEOS) and
15 triethanolamine $\geq 99\%$ (TEA) from Fluka (www.lab-honeywell.com, Bucharest, Romania).

16 The reference extraction procedure (Khan et al., 2020) was reproduced using 500 mg C₁₈
17 Varian Bond Elut cartridges from Agilent Technologies (www.agilent.com, Santa Clara, USA).

18 **2.2. Material, instrumentation, and analytical conditions**

19 During the synthesis process, cyclodextrins' drying was carried out by using an Azbil Telstar
20 LyoAlfa 10/15 freeze drier (www.azbil.com, Shonan, Japan). A rotary evaporator from Büchi
21 (www.buchi.com, Flawil, Switzerland) was also used for solvent elimination.

22 The characterization of the solid phases was done using several techniques. Transmission
23 electron microscopy (TEM) images of the materials were obtained using a Jeol microscope
24 (www.jeol.com.jp, Tokyo, Japan) model JEM-1010 operating at 200 kV. ¹³C and ²⁹Si nuclear
25 magnetic resonance (NMR) spectra were acquired at 128.3 MHz and 79.5 MHz, respectively, in
26 magic-angle spinning mode with a Varian Unity 300 spectrometer from Agilent Technologies
27 (www.agilent.com, Santa Clara, USA). X-ray diffractograms (XRD) were obtained using Cu K α

1 radiation (Seifert 3000TT θ - θ). Elemental CNH analysis was performed with an elemental
2 analyzer 1100 of CE Instruments Ltd. (www.ceinstruments.co.uk, Hindley Green, United
3 Kingdom). Porosity data were determined by using N₂ adsorption-desorption isotherms recorded
4 at 77 K with a Micromeritics ASAP2010 automated sorption analyzer (www.micromeritics.com,
5 Norcross, USA). To end, thermogravimetric analysis (TGA) was operated with a SETARAM
6 SETSYS 16-18 (www.setaram.com, Cailure-et-Cuire, France).

7 For the SPE, a Vac Elut 20 from Agilent Technologies (www.agilent.com, Santa Clara, USA)
8 using 6 mL cartridges from Análisis Vínicos (www.analisisvinicos.com, Ciudad Real, Spain) was
9 employed. All samples were previously filtered with Nylon 0.45 mm Sartorius Stedim Biotech
10 filters (www.sartorius.com, Göttingen, Germany). Also, a miVac sample concentrator from SP
11 Scientific (www.spscientific.com, Warmister, USA) was utilized for solvent evaporation.

12 For the analytes' separation and quantitation, an LC-2000 Plus liquid chromatograph equipped
13 with an FP-2020 Plus Intelligent Fluorescent Detector, a PU-2089 Plus Quaternary Gradient
14 Pump with integrated degasser, and an I/FLC/NetII/ADC interface from Jasco
15 (www.jascoinc.com, Madrid, Spain) were used. The injection volume was 20 μ L in a Supelco
16 Analytical (www.sigmaaldrich.com, Bellefonte, USA) six-way injection valve from Agilent
17 Technologies (www.agilent.com, Santa Clara, USA), and the stationary phase used was a
18 Kromasil C18 column (150 x 4.6 mm, 5 μ m particle size) from Análisis Vínicos
19 (www.analisisvinicos.com, Ciudad Real, Spain). The separation was carried out at 1 mL min⁻¹
20 using a 55:45 ACN:H₂O isocratic-mode mobile phase. To end, the fluorimetric detection was done
21 at a λ_{ex} = 254 nm and a λ_{em} = 310 nm.

22 **2.3.Preparation of cyclodextrin-modified mesoporous silica materials**

23 The first step of the synthesis corresponds to the CD silanization to make the covalent
24 attachment to the silica structure possible. The functionalization procedure has been already
25 described (Belenguer-Sapiña et al., 2020, 2018). In short, 9.4 mmol of previously dried CD (10.7
26 g for β -CD and 12.2 g for γ -CD, respectively) were mixed with 9.3 mL of 3-(triethoxysilyl)propyl
27 isocyanate in 130 mL of dry pyridine. The solution was stirred at 70 °C for 48 h. Then, the solvent

1 was removed under pressure and the solid obtained was washed thoroughly with ethanol to
2 remove pyridine traces.

3 The synthesis of the final UVM-7-type solid phases was adapted from the one used for
4 obtaining UVM-7 materials elsewhere (Pellicer-Castell et al., 2018) through an experimental
5 procedure in the presence of surfactant. The basic ingredients were mixed in the molar ratio 2
6 TEOS:7 TEA:0.5 CTAB:180 H₂O. In this case, the incorporation of cyclodextrins was carried out
7 through the previous solution of the silanized ones in the aqueous part of the reagents with a
8 molar ratio of 2% with respect to the TEOS content. Thus, the molar ratio should be 1.98
9 TEOS:0.02 CD:7 TEA:0.5 CTAB: 180 H₂O. The use of the atrane method (El Haskouri et al.,
10 2008) implies that all reagents (both the silica source and other inorganic heteroelements or
11 organically modified silanes) are forming atrane or pseudoatrane complexes before starting the
12 cohydrolysis and cocondensation processes. Thus, all reagents are supposed to be first
13 dissolved in TEA. However, the impossibility of dissolving the functionalized γ -CD in the TEA
14 medium made it necessary to modify the general method in this case. The dissolution of both
15 CDs (β - and γ -) was carried out in an aqueous medium, and they were incorporated into the
16 synthesis in the last stage. Although functionalized β -CD appeared to be soluble in TEA medium
17 (Pellicer-Castell et al., 2021), both CDs were dissolved in aqueous media in order to have
18 homogeneity in the preparation of the materials here tested.

19 Briefly, 50 mL of TEA were added to 23.7 mL of TEOS. The mixture was heated to 140 °C.
20 After a short period of cooling, 9.8 g of CTAB were added to the mixture at 120 °C and 1.8 mmol
21 of functionalized cyclodextrin (2.3 g of β -CD and 2.6 g of γ -CD, respectively) dissolved in 180
22 mL of water were slowly mixed with them under vigorous stirring. The dissolution and
23 incorporation of CDs to the rest of the reagents was carried out in very short times and under
24 vigorous stirring to avoid self-condensation processes. The mixture was aged at room
25 temperature overnight and the resulting powder was afterward filtered and washed. Then, the
26 material was dried with vacuum. To end, acid extraction was used to remove the surfactant. To
27 this end, 100 mL of ethanol and 8.5 mL of HCl 37% were added for each gram of the obtained
28 material and mixed under stirring at 60 °C.

2.4. Extraction protocol and sample analysis

The experimental procedure was optimized by studying the parameters that can influence the concentration factor obtained through SPE. Thus, they were evaluated and optimized independently, that is, by varying one parameter at a time while keeping the others constant. The optimization study was carried out based on the recovery obtained from spiked samples at the $\mu\text{g L}^{-1}$ level and the recovery was calculated from the ratio between the obtained concentrations and the theoretical concentration ($n=3$). The studied parameters were the type of adsorbent, ionic strength, pH, influence of the washing step, amount of solid phase, sample volume, loading capacity, the volume of extracting solvent, and the possibility of concentrating the elution by evaporation and later reconstitution. Besides, the chance of reusing the materials was tested. To end, the existing matrix effect was assessed by spiking juice samples to establish the variation of the recovery and the coefficient of variance of standard line slopes coming from standard addition calibration solutions. A study of CDs interfering in the determination when present in juices as food additives was also carried out.

Extraction cartridges were prepared by packing the CD-UVM-7 material between two polyethylene frits into 6 mL empty cartridges. The recommended SPE procedure consists of using 200 mg of β -CD-UVM-7 solid phase and conditioning it with 3 mL of MeOH followed by 5 mL of ultrapure water. Then, 25 mL of sample, whose ionic strength and pH are previously adjusted to 4 mol L^{-1} and pH 2.5, respectively, are passed through the cartridge at reduced pressure. The cartridges are subsequently washed with 5 mL of ultrapure water at the same pH and ionic strength as the samples and dried for 10 min. In the end, analytes are eluted with 2 mL of MeOH. After their concentration by evaporation and subsequent recomposition with 500 μL of MeOH, the extracts are analyzed by HPLC-FD.

Regarding the analytical figures of merit, linearity, sensitivity, and precision of the final protocol were established. The limit of the detection (LOD) and the limit of quantification (LOQ) were estimated following the IUPAC recommendations (Olivieri et al., 2006). The LOQ was considered as the lower limit of linearity. Besides, the precision was evaluated by studying the intra-day and inter-day repeatability through the calculation of the relative standard deviation

1 (RSD). The intra-day precision was determined by analyzing five replicates within the same day,
2 whereas the inter-day precision was established by analyzing three series of three independent
3 extractions carried out at three different times.

4 The studied EDCs were determined in five commercial apple juices sold in PET containers
5 (M1 and M2), Tetra Pack packages (M3 and M4), and glass bottles (M5) in Spanish
6 supermarkets. As mentioned, the results obtained were validated with those obtained using a
7 reference method in the literature (Khan et al., 2020). All samples were previously homogenized,
8 filtered, and pre-treated following the recommended procedure in each case. After assessing
9 the existing matrix effect, results were calculated by using a standard addition calibration.

10 **3. RESULTS AND DISCUSSION**

11 **3.1. Characterization of the mesoporous silica-based materials**

12 First, TEM images were obtained for the CD-UVM-7 materials synthesized. In both cases,
13 the incorporation of cyclodextrin units in the UVM-7 structure did not modify the typical
14 architecture of the pure mesoporous silica (Pellicer-Castell et al., 2019) studied. Representative
15 images are shown in *Figure 1*. The characteristic UVM-7 structure includes a pore system
16 combining large interparticle macropores with intra-particle large pores showing a hexagonal
17 disordered array. In the micrographs shown, the continuous silica network constructed from
18 aggregated small mesoporous nanoparticles is appreciated. It is observed that, in the case of
19 the β -CD-UVM-7 (*Figure 1a*), there has been a greater aggregation among primary
20 nanoparticles, whose clusters show, in consequence, a slightly larger size when compared to
21 γ -CD-UVM-7 (*Figure 1b*). In any case, it can be concluded that the modification of the synthesis
22 method has not prevented the obtaining of materials with UVM-7 type architecture according to
23 the obtained TEM images. However, it should be mentioned that a certain loss of organization
24 and homogeneity is observed compared to similar materials containing β -CD obtained from the
25 classical method, that is, dissolving β -CD in TEA. Despite this, TEM images show that a highly
26 accessible silica network is preserved, which is the key feature for the application described in
27 this paper.

1 The XRD patterns measured evidenced a broad signal at low angles (2° (2θ)), which is in
2 concordance with the expected intranoparticle organization based on a mesopore array with
3 hexagonal pseudo-order. In both materials, this XRD signal (*Figure S2*) is wider and of lower
4 intensity than that observed in materials prepared by the standard procedure using TEA as
5 solvent. This feature suggests that the co-condensation process starting from the silica-based
6 reagents in water does not allow such an ordered and effective self-assembly with the surfactant
7 micelles when compared to the standard synthesis starting from TEA. This trend is even more
8 pronounced in the case of the sample containing γ -CD, probably due to its larger size, which
9 can generate greater steric hindrances during the mesostructure formation. In fact, in the case
10 of the γ -CD-UVM-7, the XRD signal is observed as a shoulder. Then, the materials obtained,
11 although clearly of UVM-7 type according to TEM images, present a slightly lower order in the
12 intraparticle mesopore system. However, it is important to note that the order degree does not
13 suppose an added value for certain applications. In this case, the important characteristic is to
14 achieve a good dispersion of active groups (β - and γ -CD) accessible for the different analytes
15 to be retained. This objective seems to be achieved according to the data presented.

16 The surface area and pore distribution of the prepared materials were determined based on
17 the N_2 adsorption-desorption isotherms. In all cases, isotherms showed two adsorption steps
18 (*Figure S3*), thus confirming their bimodal framework and high accessibility (Pellicer-Castell et
19 al., 2018). Data in *Table 1* is besides consistent with TEM and XRD observations. While in the
20 case of the β -CD material the intraparticle porosity is significant ($0.34 \text{ cm}^3 \text{ g}^{-1}$), it decreases for
21 the material with γ -CD ($0.14 \text{ cm}^3 \text{ g}^{-1}$). Oppositely, the interparticle textural porosity is higher for
22 the γ -CD-UVM-7 ($0.71 \text{ cm}^3 \text{ g}^{-1}$) when compared to the β -CD-UVM-7 ($0.25 \text{ cm}^3 \text{ g}^{-1}$). The lower
23 particle aggregation in the latter material also correlates with its larger textural pore size. On the
24 other hand, the greater size of the intraparticle mesopore observed for the γ -CD-UVM-7 material
25 is probably related to the greater size of the cyclodextrin and the steric problems it may induce
26 in the self-assembly process among silica and the micelles. The preserved porosity is useful
27 enough for EDCs trapping in the CD active centers of the solid phases within the extracting
28 conditions tested and can contribute to higher selectivity of the analytical methodology applied.

1 In addition, the amount of CD incorporated onto the silica network described was estimated
2 from the TGA curves obtained (*Figure S4*). In both cases, a first weight loss at $T < 90^{\circ}\text{C}$ can be
3 attributed to hydration water molecules of the phases. At higher temperatures ($200\text{-}500^{\circ}\text{C}$), the
4 relatively abrupt weight loss observed in both cases can be associated with the degradation and
5 oxidation of cyclodextrins. Later, a less pronounced weight loss due to the silanol condensation
6 occurs leading to the SiO_2 final residue. The weight loss values associated with the cyclodextrins
7 decomposition are 21 and 23% for $\beta\text{-CD-UVM-7}$ and $\gamma\text{-CD-UVM-7}$, respectively. Considering
8 that the number of silanol arms attached to both cyclodextrins is very similar, ca. 4 (3.8 and 3.9
9 for the $\beta\text{-CD}$ and the $\gamma\text{-CD}$, respectively (Belenguer-Sapiña et al., 2018)), it is possible to
10 estimate the functionalization degree assuming a simplified $\text{CD}_x\text{-SiO}_2$ formulation. An identical
11 value of $x=0.01$ is obtained for both materials, which corresponds to a degree of CD
12 incorporation of 0.13 mmol/g (corresponding to C wt% values of 9.7 and 10.4 for $\beta\text{-CD-UVM-7}$
13 and $\gamma\text{-CD-UVM-7}$, respectively). These values are in excellent agreement with the elemental
14 analysis data; 9.9 C wt% ($\beta\text{-CD-UVM-7}$) and 10.6 C wt% ($\gamma\text{-CD-UVM-7}$). Due to the low nitrogen
15 content of the modified CD structures, the C content was used as the most reliable data to
16 determine the concentration of CD in our materials. Important is to mention that results of the
17 TGA experiments also permitted intuiting fine stability of the solid phase up to temperatures of
18 at least 200°C , which is in accordance with previous studies on type UVM-7 materials (El
19 Haskouri et al., 2002).

20 Regarding the NMR experiments, the ^{29}Si and ^{13}C NMR spectra of the solid phases were
21 recorded. Results are shown in *Figure S5* and they were comparable for both materials. As can
22 be seen in *Figure S5a*, the ^{29}Si NMR of the modified CD (either $\beta\text{-}$ or $\gamma\text{-}$) showed a unique signal,
23 which is compatible with the presence of silicon T^2 and T^3 sites presence corresponding to their
24 structure. Oppositely, the final materials evidenced the existence of a minority of T silicon sites
25 in comparison with the new presence of Q^2 , Q^3 , and Q^4 silicon sites (*Figure S5b*), also coinciding
26 with the expected molecular structure of the final material. The ^{13}C NMR spectra supported this
27 information. In this case, both spectra were very similar. However, the one corresponding to the
28 modified CDs (*Figure S5c*) showed two extra peaks in comparison with the spectra of the final
29 material (*Figure S5d*). This is consistent with the disappearance of the ethyl groups in the

1 synthesis of the final solid phase due to the covalent attachment of the modified cyclodextrin to
2 the silica structure through its silanol groups.

3 **3.2. Influence of the CD type in the retention**

4 The first step of the optimization study consisted of assessing the influence of the sorbent
5 nature on the extraction performance. This study was carried out by testing either pure UVM-7
6 silica as well as the same type of mesoporous silica doped with β -CD and γ -CD (β -CD-UVM-7
7 and γ -CD-UVM-7, respectively). To this, SPE cartridges containing 100 mg of solid phase were
8 constructed and 10 mL of ultrapure water spiked with 50 $\mu\text{g L}^{-1}$ of each of the target analytes
9 were loaded into the cartridges. The elution was accomplished with 3 mL of MeOH. Besides,
10 water was collected after passing through the cartridge to be able to check the retention together
11 with the extraction performance achieved. The best results (*Figure 2*) were obtained when using
12 the β -CD-UVM-7 solid phase in all cases. In the case of parabens, recoveries were of 72, 100,
13 and 108% for MP, EP, and PP, respectively. In the case of bisphenols, all recoveries were
14 around 100% for BPA, BPAP, and BPC. Thus, results obtained strongly support the idea of the
15 size of CDs influencing the formation of the inclusion complexes with the analytes (Belenguer-
16 Sapiña et al., 2018, 2021). For parabens, better recoveries were observed as the analytes'
17 molecular size increases (*Table S1*). As a shift from methyl (size 0.74 nm) to propyl-substituents
18 (size 0.86 nm) is carried out, recoveries improve significantly due to a better fit between
19 parabens and β -CD (average β -CD cavity size is 0.80 nm). The same behavior was observed
20 in the case of UVM-7, probably due to a slight contribution of the silica mesopores in the
21 retention, as well as in the case of γ -UVM-7, whose internal cavity is possibly too large for them
22 (average γ -CD cavity size is 0.90 nm). In this case, analytes may be able to enter inside γ -CD,
23 but the interactions taking place are not strong enough to ensure proper retention. The case of
24 bisphenols is not entirely comparable. As shown, β -CD-UVM-7 worked better with them in all
25 cases for the same reason as in the previous case. However, no differences were observed for
26 the three analytes, probably due to their similarities at a structural level, which lead to thinking
27 that the structural differences between them should not be as significant as in the case of
28 parabens (*Table S1*). In any case, these results also support the influence of CD on the retention

1 mechanism taking place, with higher recoveries being obtained in all cases with the presence
2 of accessible cyclodextrin units onto the silica structure.

3 **3.3. Optimization of the extraction conditions**

4 First, the elution step was studied. Methanol was chosen as extracting solvent due to the
5 quantitative recoveries with respect to the amount of analytes retained in the solid phase already
6 observed. Moreover, its toxicity is lower than that of other solvents that may be tested, such as
7 acetonitrile. The elution profile of methanol was thus assessed to use the minimum but enough
8 amount of solvent in the optimized procedure. Different extractions were carried out by using
9 volumes of 1, 2, 3, and 4 mL of methanol for the elution process. The results obtained showed
10 that the elution was complete with only 2 mL of methanol. Concretely, 1 mL of methanol
11 extracted from 60% to 90% of the studied compounds, depending on the analyte observed,
12 while the second milliliter eluted the rest (from 10% to 35% in addition to the previous amount).
13 No analytes were extracted in the next third milliliter.

14 Also, the possibility of concentrating the extract by evaporation and subsequent
15 reconstitution was studied. The treatment was carried out under vacuum at 60 °C for 15 min
16 followed by redissolution with 500 µL of methanol. Variations lower than 4% were obtained in
17 the final recovery, probably due to the high stability these analytes show to temperature (*Table*
18 *S1*). Thus, the concentration step is recommended to increase the sensitivity of the method.

19 Regarding the loading conditions, the first parameter studied was the pH. As it is known, fruit
20 juices usually present an acidic pH. Due to the acid-base properties that the compounds under
21 study may present (*Table S1*), together with the influence of pH on the encapsulation properties
22 of cyclodextrin and the common acidic pHs (ca. pH 3) found in fruit juices, it was decided to test
23 the use of loading pH 2.5 in comparison to the neutral pH used so far. The mentioned value was
24 adjusted by dropping concentrated H₂SO₄ into the spiked sample. For MP and EP, the recovery
25 improved at a rate in the range 5 – 24% from neutral pH to the acidic pH tested. The rest of the
26 analytes maintained the already shown quantitative recoveries. These results reinforce the idea
27 of obtaining good extraction results in the analysis of real juice samples.

Besides, the ionic strength was evaluated through the modification of the NaCl content (from 0 mol L⁻¹ to 4 mol L⁻¹). Recoveries improved considerably as the ionic strength increased to the best results with the use of 4 mol L⁻¹. The recoveries, in this case, were 87%, 102%, 100%, 98%, 93%, and 92 % for MP, EP, BPA, PP, BPAP, and BPC, respectively.

Other analytical parameters such as the breakthrough volume were also assessed. To this, several spiked samples were treated following the optimized SPE procedure, varying both the sample volume and the amount of solid phase each time. Results showed that good recoveries were obtained when using 100 mg of solid phase to extract 10 mL of sample. In order to improve the sensitivity, 25 mL and 50 mL of sample were then tested. However, significant losses were quantified in this case. For this reason, 200 mg of solid phase were also tested to extract 25 mL and 50 mL of sample. Results indicate that either 25 mL or 50 mL of sample can be treated by using 200 mg of the developed material, being 25 mL volume acceptable for the treatment of juice while achieving a good sensitivity.

The loading capacity was tested with spiked samples from the ng L⁻¹ level to the µg L⁻¹ level. No higher concentrations were assessed due to the expected low content of the studied compounds in the selected samples. The only little variations observed in the recoveries obtained for the different analytes in the range studied can be simply due to the dispersion of the results and not to the influence of the concentration.

To conclude, the reusability of the material was studied. Results show that the solid phase is reusable for at least four extractions by maintaining the recovery. For this reason, the possibility of reusing the material can be included among its functionalities and constitutes therefore a great advantage. These results were expected due to those obtained in previous studies when using polymeric materials based on cyclodextrins (Belenguer-Sapiña et al., 2019).

3.4. Analytical figures of merit

The analytical performance of the optimized SPE protocol was evaluated concerning global recovery, precision, limits of detection (LODs), and limits of quantification (LOQs), and linearity. Parameters informed in *Table 2* were estimated according to the recommendations of the

1 IUPAC (Olivieri et al., 2006). Concretely, the lower limit of the linearity range was established
2 from the LOQs.

3 As mentioned, the precision of the method was evaluated by measuring the intra-day and
4 inter-day repeatability from different extractions of spiked samples and was expressed in terms
5 of the obtained RSD. The intra-day precision was estimated by analyzing three replicates within
6 a day, whereas the inter-day precision was calculated by analyzing three series of three
7 experiments carried out on three days. As can be seen, the RSDs range from 0.5% in the lower
8 limit to 6.8%. In any case, the method showed good precision with deviation values below 7%.

9 Besides, both LOD and LOQ were calculated referred to the initial sample by considering the
10 concentration factor provided by the described extraction procedure. LODs are calculated
11 between 16 and 28 ng L⁻¹ and LOQs are below 86 ng L⁻¹, showing the high sensitivity obtained
12 when using the proposed method. The experimental procedure developed together with the
13 detection method selected provide a satisfactory sensitivity. This would be expected since
14 fluorescence detection is usually characterized for showing high selectivity and sensitivity.

15 Also, overall recoveries greater than 90% were achieved for all analytes under study, which
16 made the concentration factors being established in the range from 46.5 to 48.5.

17 An acceptable linearity range permits the use of the developed extraction methodology in a
18 range of concentrations of the measuring solution.

19 To end, *Table S2* compares the characteristic analytical features of the developed method
20 with other methodologies reported in the literature for the determination of EDCs in different
21 types of samples. Thus, it can be stated that the here-presented analytical parameters obtained
22 were at least comparable or even better than those commonly found in the literature
23 (Asimakopoulos et al., 2014; Azzouz & Ballesteros, 2016; Chang et al., 2012; Chormey, Bodur,
24 Baskin, Firat, & Bakirdere, 2018; Ferrer et al., 2011; Mousa et al., 2013; Vela-Soria et al., 2013).
25 As can be seen, many of the reported methods use extraction systems that do not involve
26 adsorbing solid phases. However, and as mentioned before, its use can help to considerably
27 improve certain characteristics of the extraction, such as the selectivity achieved. This behavior
28 can be clearly seen, given that the lowest detection limits and variation of the recoveries

1 obtained are accomplished when the mentioned adsorbents are used. Furthermore, the usage
2 of reusable materials results in greener and more sustainable analytical methodologies.

3 **3.5. Matrix influence**

4 The matrix effect was studied in a two-way process. First, the extraction efficiency of real
5 juice samples spiked with $50 \mu\text{g L}^{-1}$ of each of the EDCs under study, following both the proposed
6 method and the chosen reference method (Khan et al., 2020), was assessed. Results are shown
7 in *Table 3*. As expected, the recovery obtained was lower than that achieved during the
8 optimization study with spiked ultrapure water due to the enhanced complexity of the matrices.
9 It is important to highlight the significantly lower recoveries obtained in the case of the reference
10 method. Concretely, EP, BPA, and PP were mostly lost following the recommended extraction
11 procedure. In the case of the proposed extraction methodology, it is important to highlight the
12 case of BPA, whose losses were minor in comparison with those observed for the rest of the
13 target compounds. Based on the results obtained, the determination of EDCs in juice samples
14 must be carried out by applying a standard addition method.

15 For this reason, a standard addition calibration was measured in comparison with the
16 external calibration used so far to confirm the matrix effect and analyze real samples. As can be
17 seen in *Figure S6*, significant variations in the slopes of the calibration curves were obtained in
18 all cases excepting the BPA, which would coincide with the behavior observed in the last
19 experiment. Moreover, it can be stated that the matrix effect was higher for MP and EP (-53%
20 and -33% of slope variation), followed by PP, BPAP, and BPC (-26%, -24%, and -20%,
21 respectively).

22 **3.6. Study of interferences**

23 Free cyclodextrins are commonly added in solution to natural fruit juices as additives to
24 guarantee long term preservation and to keep their good properties (Matencio, Navarro-
25 Orcajada, García-Carmona, & López-Nicolás, 2020; Pereira et al., 2021). Since their presence
26 may constitute potential interfering substances in the determination of EDCs developed through
27 maintaining them in solution and avoiding their retention in the solid phase, a study of

interferences was carried out to evaluate this possibility. To it, the extraction efficiency of real juice samples spiked with 50 $\mu\text{g L}^{-1}$ of each of the EDCs under study containing a 15 mM concentration of either free α -, β -, or γ -CD in solution (Andreu-Sevilla, López-Nicolás, Carbonell-Barrachina, & García-Carmona, 2011) was evaluated. Results showed that free CDs represent indeed potential interfering ingredients. Specifically, α -CD potentially interfered with a decrease in the recoveries for MP, EP, and PP (recoveries decreased to 10%, 12%, and 7%, respectively). In the same way, β -CD interfered for MP, EP, PP, BPC, and slightly for BPA and BPAP (recoveries quantified were of 12%, 13%, 60% 6%, 11%, and 55% for MP, EP, BPA, PP, BPAP, and BPC, respectively). Finally, γ -CD showed to disturb in the recovery for BPA, BPAP, BPC, and slightly for PP (recoveries decreased to 21%, 25%, 18%, and 40%, respectively), while the rest of analytes got their initial values back. The explanation to these phenomena may be found in the sizes of the CD cavities together with the sizes of the analytes molecules (*Table S1*), as commented previously. This is, analytes that fit better with each CD decreased the recovery obtained in the solid phase since they maintained in solution, while the rest were able to be quantified in the same conditions.

3.7. Application to real juice samples

At this point, the target EDCs were determined in five commercial juice samples (*Table 4*) free of CDs in solution. Given them, it can be concluded that EDCs are present in juice at trace level, as expected. The compounds that are more commonly found in the packages studied are MP, PP, and BPA. Moreover, PET containers show an increased content of the variety of EDCs in comparison with, for example, Tetra Pack containers. The presence of PP, BPAP, and BPC is less remarkable.

Regarding the comparison of the results with those obtained using the reference extraction methodology, it can be stated that both methods are comparable among them (Harris, 2007). The differences observed in the results may be due to (a) the interferences in the reference method, since the solid phase used does not present a high selectivity and this fact may influence the detection step, and (b) the higher detection limits of the reference method coming from the significantly lower extraction efficiencies observed, which could make a β -CD-UVM-7-

1 detectable compound non-detectable. However, it can be observed that the results obtained in
2 both cases are at the same concentration level, that is, at the trace level, so the method
3 developed can be considered validated.

4 **4. CONCLUSIONS**

5 In this study, a type UVM-7 mesoporous silica material containing cyclodextrin as surface
6 modifier has been synthesized. The morphology and properties of the newly developed support
7 have been entirely characterized. The material is sufficiently stable and has demonstrated to be
8 reusable, constituting both properties great advantages in comparison with other disposable
9 commercial adsorbents. Then, the extracting features of the developed sorbent have been
10 validated through the SPE of endocrine-disrupting chemicals from acidic fruit juices for their
11 subsequent quantification.

12 The positive influence of cyclodextrins in the structure of the material on the adsorption
13 mechanism carried out has been verified. Moreover, the idea of adapting the synthesis of the
14 base material to the type of analytes in use taking into account the influence of the cyclodextrin
15 size on the retention of different families of compounds according to their physicochemical
16 properties (Belenguer-Sapiña et al., 2018, 2021) is supported here. Cyclodextrins certainly
17 contribute to the matrix clean-up and the selectivity-enhanced extraction of the target
18 compounds prior to their analysis. As a disadvantage, other ingredients of the sample, such as
19 free CDs present as additives, may result in interfering substances for the determination
20 described and may be taken into account.

21 After the optimization of the extraction parameters, good sensitivity has been achieved and
22 the repeatability of the method can be classified as satisfactory. The reported method represents
23 a simple procedure as a promising tool for systematically analyzing migrating EDCs residues in
24 bottled acidic fruit juices. As a future insight, the use of the developed sorbent to clean fruit
25 juices from high-concern substances as EDCs prior to consumption may also be a topic of
26 interest to study.

27 **CONFLICT OF INTEREST**

28 The authors declare no conflict of interest.

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4

1 TABLES

2 **Table 1.** Physical and textural parameters of the CD-UVM-7 solid phases tested.

Solid phase	BET surface area (m ² g ⁻¹)	Mesopores		Large pores	
		Pore size (nm)	Pore volume (cm ³ g ⁻¹)	Pore size (nm)	Pore volume (cm ³ g ⁻¹)
β-CD-UVM-7	686 ± 1	2.4 ± 0.3	0.34 ± 0.03	30.1 ± 0.5	0.25 ± 0.04
γ-CD-UVM-7	353 ± 2	2.8 ± 0.4	0.14 ± 0.03	41.4 ± 0.6	0.71 ± 0.05

3

1 **Table 2.** Analytical figures of merit of the method developed.

Compound	RSD (%) Intra-day	RSD (%) Inter-day	Recovery (%)	LOD ^a (ng L ⁻¹)	LOQ ^a (ng L ⁻¹)	Linearity ^b (µg L ⁻¹)
MP	3.9	0.5	93 ± 4	26 ± 5	80 ± 15	2.5 - ≥ 500
EP	2.2	0.4	96 ± 3	28 ± 7	86 ± 20	2.7 - ≥ 500
BPA	4.6	3.3	95 ± 4	26 ± 2	78 ± 4	2.4 - ≥ 500
PP	4.7	1.8	97 ± 3	24 ± 2	73 ± 4	2.3 - ≥ 500
BPAP	6.8	6.4	94 ± 5	16.0 ± 0.8	50 ± 3	1.5 - ≥ 500
BPC	2.2	2.9	97 ± 3	17.0 ± 1.4	52 ± 4	1.6 - ≥ 500

2 ^a LOD and LOQ are referred to the sample under treatment.

3 ^b Linearity is referred to the measuring solution.

1 **Table 3.** Influence of the matrix on the recovery of EDCs from apple juice.

Compound	Recovery (%)	
	β -CD-UVM-7	C18 (Khan et al., 2020)
MP	63 $\pm$ 3	48.5 $\pm$ 1.1
EP	69 $\pm$ 3	23 $\pm$ 5
BPA	80.0 $\pm$ 1.8	21 $\pm$ 4
PP	59 $\pm$ 3	11 $\pm$ 2
BPAP	64 $\pm$ 7	58 $\pm$ 2
BPC	64 $\pm$ 5	59 $\pm$ 2

2

1 **Table 4.** Analysis of EDCs in apple juice by comparing a reference method (Khan et al., 2020) and the developed method ($\mu\text{g L}^{-1}$, $\bar{x} \pm s$, $n=3$).

Compound	M1		M2		M3		M4		M5	
	PET container		PET container		Tetra Pack package		Tetra Pack package		Glass bottle	
	$\beta\text{-CD-UVM-7}$	C_{18}	$\beta\text{-CD-UVM-7}$	C_{18}	$\beta\text{-CD-UVM-7}$	C_{18}	$\beta\text{-CD-UVM-7}$	C_{18}	$\beta\text{-CD-UVM-7}$	C_{18}
MP	1.92 ± 0.15	1.8 ± 0.1	16.2 ± 0.7	14 ± 3	56.0 ± 0.1	44 ± 3	2.5 ± 0.3	< LOD	< LOD	< LOD
EP	< LOD	1.2 ± 0.1	< LOD	< LOD	74 ± 2	68 ± 7	10 ± 2	11.0 ± 0.3	8.1 ± 0.4	8.9 ± 0.4
BPA	7.8 ± 0.4	< LOD	12.8 ± 0.8	< LOD	5 ± 1.3	< LOD	< LOD	< LOD	1.5 ± 0.3	< LOD
PP	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD
BPAP	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD
BPC	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD	< LOD

2

1 **FIGURE CAPTIONS**

2 **Fig.1** Representative TEM images of (a) β -CD-UVM-7 and (b) γ -CD-UVM-7 materials
3 showing the typical architecture of the UVM-7 derivatives.

4 **Fig.2** Influence of the nature of the sorbent used on the recovery of EDCs: **(a)** parabens, and
5 **(b)** bisphenols.



